HR Analytics Dashboard using Microsoft Excel

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Tools Used: Microsoft Excel

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1. Introduction

Human Resource (HR) Analytics is the process of collecting, organizing, and analyzing employee-related data to improve business decisions. Organizations use HR Analytics to understand employee performance, identify attrition patterns, improve employee retention, and enhance workforce planning.

This project focuses on analyzing employee data using Microsoft Excel and creating an interactive HR Analytics Dashboard. The dashboard helps HR managers monitor important employee metrics through KPIs, Pivot Tables, Pivot Charts, and Slicers.

2. Project Objectives

The main objectives of this project are:

- Analyze employee information using Microsoft Excel.
- Calculate important HR Key Performance Indicators (KPIs).
- Identify employee attrition patterns.
- Compare attrition across departments and job roles.
- Create an interactive HR Analytics Dashboard.
- Generate business insights for HR decision-making.

3. Dataset Information

Dataset Name: IBM HR Analytics Employee Attrition & Performance

Dataset Type: Employee Records

Number of Employees: 1470

Number of Attributes: 35

Important Columns

- Employee Number
- Age
- Gender
- Department
- Job Role
- Monthly Income
- Education Field
- Years at Company
- Attrition
- Job Satisfaction
- Performance Rating
- Business Travel

4. Tools & Technologies

The following tools were used:

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Excel Formulas
- Slicers
- Dashboard Design

5. Data Cleaning

Before performing the analysis, the dataset was cleaned and prepared.

The following preprocessing steps were completed:

- Converted the dataset into an Excel Table.
- Checked for missing values.
- Verified data consistency.
- Applied proper formatting.
- Organized the dataset for Pivot Table analysis.

6. Dashboard Development

The dashboard was created using Microsoft Excel.

The following features were included:

- KPI Cards
- Pivot Tables
- Pivot Charts
- Interactive Slicers
- Dashboard Layout

The dashboard allows users to filter data dynamically and analyze employee information efficiently.

7. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Total Employees
- Attrition Count
- Attrition Rate
- Retention Rate
- Average Employee Age
- Average Monthly Income

These KPIs provide a quick overview of workforce statistics.

Cell	Value
HR Analytics Dashboard	
Total Employees	1470
Attrition Count	237
Attriition Rate	16%
Retention Rate	84%
Average Age	36.92381
Average Monthly Income	6502.931

Key Performance Indicators (KPIs) of the HR Analytics Dashboard

8. Dashboard Visualizations

The following visualizations were created:

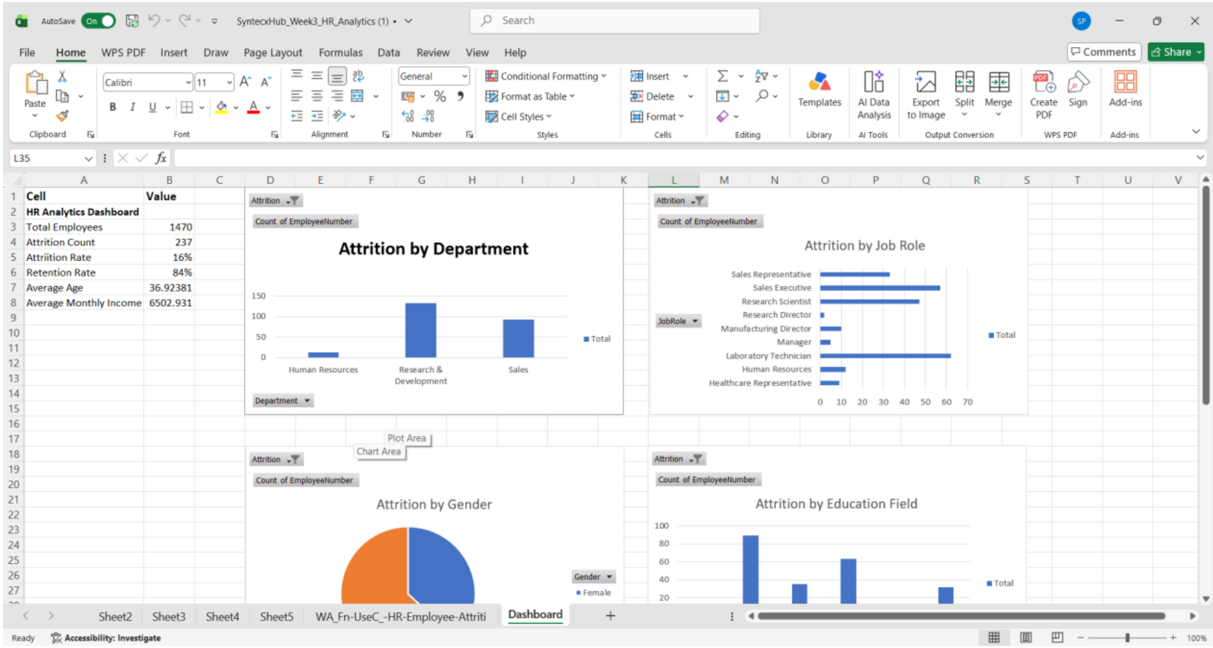
- Attrition by Department
- Attrition by Job Role
- Attrition by Gender
- Attrition by Education Field

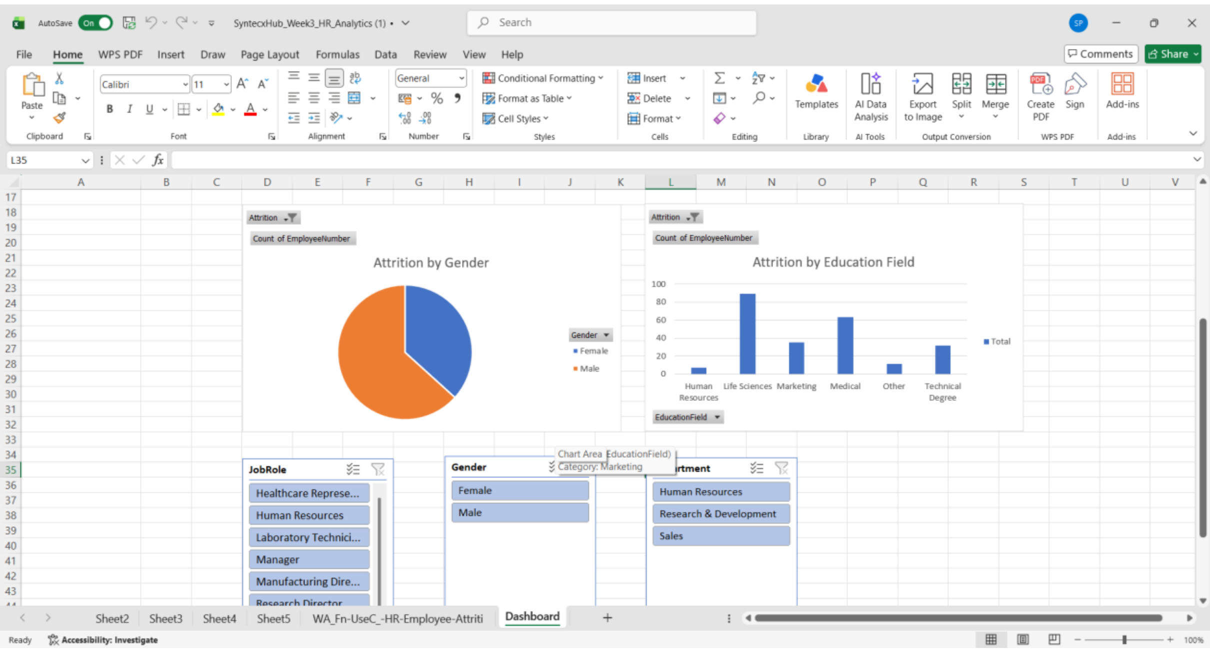
Interactive slicers were added for:

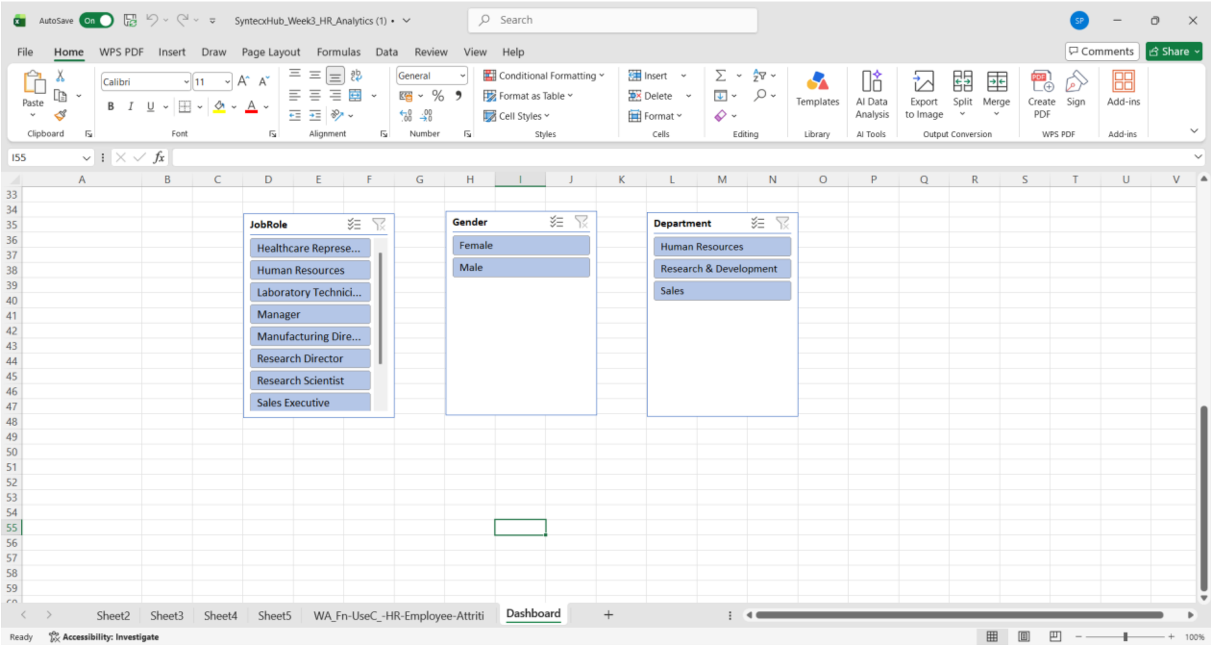
- Department
- Gender

- Job Role

These visualizations help HR managers compare employee attrition across different categories.







9. Business Insights

The dashboard generated the following insights:

- Total employees analyzed: 1470
- Total attrition count: 237
- Overall attrition rate: 16.12%
- Employee retention rate: 83.88%
- Research & Development experienced the highest employee attrition.
- Certain job roles showed higher employee turnover than others.
- The average employee age is approximately 37 years.
- The average monthly income is approximately 6503.
- Interactive slicers allow detailed analysis based on departments, gender, and job roles.

10. Conclusion

This project demonstrates how Microsoft Excel can be effectively used for HR Analytics. The dashboard provides an interactive and user-friendly interface for analyzing employee data, monitoring workforce trends, and identifying areas with higher attrition.

The project helps HR professionals make data-driven decisions that improve employee retention and workforce planning.

11. Future Enhancements

Future improvements include:

- Develop the dashboard using Microsoft Power BI.
- Add predictive analytics for employee attrition.
- Integrate real-time employee data.
- Create advanced visualizations and forecasting.
- Automate dashboard updates using Power Query.

12. References

- IBM HR Analytics Employee Attrition & Performance Dataset
- Microsoft Excel Documentation
- Microsoft PivotTable Documentation
- Microsoft PivotChart Documentation